

Problem

Suppose your circuit laboratory has the following standard commercially available resistors in large quantities:

$$1.8\ \Omega \quad 20\ \Omega \quad 300\ \Omega \quad 24\ \text{k}\Omega \quad 56\ \text{k}\Omega$$

Using series and parallel combinations and a minimum number of available resistors, how would you obtain the following resistances for an electronic circuit design?

- (a) $5\ \Omega$
- (b) $311.8\ \Omega$
- (c) $40\ \text{k}\Omega$
- (d) $52.32\ \text{k}\Omega$

Step-by-step solution

Step 1 of 4

(a)

Write the possible combinations of resistances to obtain a $5\ \Omega$ resistance.

$$\begin{aligned} 5\ \Omega &= (10 \parallel 10)\ \Omega \\ &= (20 \parallel 20) \parallel (20 \parallel 20)\ \Omega \end{aligned}$$

Hence, four resistors of $20\ \Omega$ in parallel are required to obtain a $5\ \Omega$ resistance.

Step 2 of 4

(b)

Write the possible combinations of resistances to obtain a $311.8\ \Omega$ resistance.

$$\begin{aligned} 311.8\ \Omega &= (300 + 10 + 1.8)\ \Omega \\ &= (300 + (20 \parallel 20) + 1.8)\ \Omega \end{aligned}$$

Hence, the series connection of a $300\ \Omega$ resistor with a parallel combination of two $20\ \Omega$ resistors and a $1.8\ \Omega$ resistor is required to obtain a $311.8\ \Omega$ resistance.

Step 3 of 4

(c)

Write the possible combinations of resistances to obtain a $40\ \text{k}\Omega$.

$$\begin{aligned} 40\ \text{k}\Omega &= (28 + 12)\ \text{k}\Omega \\ &= [(56 \parallel 56) + (24 \parallel 24)]\ \text{k}\Omega \end{aligned}$$

Hence, the series connection of the parallel combination of two resistors of $56\ \text{k}\Omega$ with a parallel combination of two $24\ \text{k}\Omega$ resistors is required to obtain a $40\ \text{k}\Omega$.

Step 4 of 4

(d)

Write the possible combinations of resistances to obtain a $52.32\text{ k}\Omega$.

$$\begin{aligned}52.32\text{ k}\Omega &= 28\text{ k}\Omega + 24\text{ k}\Omega + 0.32\text{ k}\Omega \\&= (56 \parallel 56)\text{ k}\Omega + 24\text{ k}\Omega + 320\Omega \\&= (56 \parallel 56)\text{ k}\Omega + 24\text{ k}\Omega + 300\Omega + 20\Omega\end{aligned}$$

Hence, a series combination of two parallel resistors of $56\text{ k}\Omega$ with a $24\text{ k}\Omega$ resistor, a 300Ω resistor and a 20Ω is required to obtain a $52.32\text{ k}\Omega$ resistor.